Pôle Méca's Internal Seminar

Tomographic shearmetry of dynamic flows at cell surfaces using nanoprobles

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Path-following methods for quasi-static crack propagation : Application to phase-field fracture

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Abstract - Arianna Giannetti

Fluid dynamic shear stress plays a crucial role in regulating numerous cellular functions. In the vascular system, abnormal endothelial responses to shear have been implicated in atherosclerosis. It is commonly assumed that vascular endothelial cells (ECs) have no impact on the flow field. Computational simulations, however, suggest that EC topography can lead to significant subcellular shear stress gradients [1]. Experimentally validating these computational predictions has remained elusive because conventional flow measurement techniques such as particle image velocimetry (PIV) are very noisy due to inaccuracies in near-wall velocity measurement [2].

$LaPO_4$:Eu nanorods suspended in a flow field align in the direction of flow, and the extent of alignment depends on the local shear stress. Thus, spectroscopic analysis of the collective orientation of the nanorods enables tomographic mapping of the local shear stress, rendering these nanorods as high-resolution shear probes [3-5]. We are exploring the use of this technique for mapping the shear stress distribution on the surfaces of ECs in vitro. In parallel, we use atomic force microscopy (AFM) and quantitative phase imaging (QPI) to obtain EC topography in order to understand how this topography impacts the shear stress distribution. We also conduct computational fluid dynamics (CFD) simulations to compute the local wall shear stress on cellular height maps.

First results on isolated ECs indicate increased nanorod alignment with flow at the cell apex compared to adjacent cell-free regions. Topographical measurements demonstrate that restricting EC spreading by culturing cells on 15 μm-wide adhesive line patterns elicits cell elongation and significant height reduction. CFD simulations reveal that surface undulations create cell-scale shear stress variations that are attenuated by cell height reduction on the line-patterned surfaces. The present results suggest that nanorod-based tomographic shearmetry may enable the first experimental mapping of shear stress distributions on EC surfaces. Future efforts will focus on correlating the local shear stress distributions with functional cellular responses such as intracellular calcium mobilization in order to elucidate the relationship between shear, cell shape and biological activity.

References

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Abstract - Flavien Loiseau

Quasi-static crack propagation simulations in brittle materials often face physical instabilities, like snapback events, due to structural softening. In variational phase-field fracture models, these instabilities cause abrupt crack jumps, undermining physical accuracy as the energy is minimized over large crack increments. They also often prevent using incremental force boundary conditions as they may lead to a loss of force balance as soon as the crack starts propagating. To tackle those issues, this study analyzes different load control methods, also called path-following methods. We evaluate existing approaches and propose a new method, the Control by Maximum Strain Increment Outside the Crack (CMSIOC), which ensures stable crack growth by limiting strain increments in uncracked regions. All methods use a single scalar control equation based on displacement fields, allowing seamless integration into classical staggered solvers. Performance is tested on three benchmark problems of increasing complexity, with results compared to Linear Elastic Fracture Mechanics (LEFM) references, including Griffith's criterion and the G-max criterion.